

Zhengyan D. Shi

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Current Position

Stanford University

Postdoctoral Fellow, *Leinweber Institute for Theoretical Physics*

Aug. 2025 – current

Education

Massachusetts Institute of Technology

Ph.D. in Physics

GPA: 5.0/5.0

Thesis title: “The Death of Quasiparticles: Strongly Interacting Gapless Phases with Fermi Surfaces and Fractional Statistics” - Advised by Prof. Senthil Todadri

Sept. 2020 – May 2025

Stanford University

B.A. in English with distinction

B.Sc. in Physics with honors and distinction

GPA: 4.07/4.0

Sept. 2016 – June 2020

Preprints and Publications

(*indicates equal contribution, α - β indicates alphabetical order)

1. **Metallogenic quantum criticality: Fermi surface nucleation at transitions between gapped phases**
[arxiv: 2608.27453](#)
Zhengyan D. Shi
2. **Skyrmion Fractional Chern Insulator: An Intrinsically Multiband Route to Fractionalization in Rhombohedral Graphene**
[arxiv: 2608.14535](#)
Julian May-Mann*, Tixuan Tan*, Patrick J. Ledwith, **Zhengyan D. Shi**, Trithip Devakul
3. **Lattice composite Fermi liquid with broken inversion symmetry**
[arxiv: 2607.28613](#)
Pavel A. Nosov and **Zhengyan D. Shi**
4. **Superconductivity and non-Fermi liquid metals in a charge-1/3 anyon fluid**
[arxiv: 2606.20403](#)
Zhengyan D. Shi and T. Senthil
5. **Charge-4e superconductor with parafermionic vortices: A path to universal topological quantum computation**
[arxiv: 2602.06963](#)
Zhengyan D. Shi*, Zhaoyu Han*, Srinivas Raghu, Ashvin Vishwanath
6. **Composite Bogoliubov Fermi liquid in a half-filled Chern band**
Phys. Rev. Lett. 137, 066604, [arxiv: 2601.09924](#)
Zhengyan D. Shi and Pavel A. Nosov

7. **Non-Abelian topological superconductivity from melting Abelian fractional Chern insulators**
[arxiv: 2512.17996](#)
Zhengyan D. Shi and T. Senthil
8. **Bidirectional Ultrafast Control of Charge Density Waves via Phase Competition**
 Phys. Rev. Lett. 135, 246504 (Editor's suggestion), [arxiv: 2510.00204](#)
 Honglie Ning, Kyoung Hun Oh, Yifan Su¹, **Zhengyan D. Shi**, Dong Wu, Qiaomei Liu, B.Q. Lv, Alfred Zong, Gyeongbo Kang et al.
9. **Charge-2e Superconductivity from a Disordered Pair Density Wave**
 Phys. Rev. Lett. 137, 116003, [arxiv: 2509.20435](#)
 Julian May-Mann, Akshat Pandey, **Zhengyan D. Shi**, Steven A. Kivelson
10. **Anyon delocalization transitions out of a disordered FQAH insulator**
 Proceedings of the National Academy of Sciences 122, e2520608122 (2025), [arxiv: 2506.02128](#)
Zhengyan D. Shi and T. Senthil
11. **Doping lattice non-abelian quantum Hall states**
 SciPost Phys. 19, 150 (2025), [arxiv: 2505.02893](#)
Zhengyan D. Shi, Carolyn Zhang, and T. Senthil
12. **Doping a fractional quantum anomalous Hall insulator**
 Phys. Rev. X 15, 031069 (2025), [arxiv: 2409.20567](#)
Zhengyan D. Shi and T. Senthil
13. **Analytic framework for self-dual criticality in $\mathbb{Z}_k$ gauge theory with matter**
 Phys. Rev. B 112, L081111 (2025), [arxiv: 2407.07941](#)
Zhengyan D. Shi and Arkya Chatterjee
14. **Excitonic quantum criticality: from bilayer graphene to narrow Chern bands**
 Phys. Rev. B 111, 125154 (2025), [arxiv:2402.12436](#)
Zhengyan D. Shi, Hart Goldman, Zhihuan Dong, and T. Senthil
15. **Controlled expansion for transport in a class of non-Fermi liquids**
 Phys. Rev. B 109, 195110 (2024), [arxiv:2311.12922](#)
Zhengyan D. Shi
16. **Local dynamics and the structure of chaotic eigenstates**
 Phys. Rev. B 108, 224305 (2023), [arxiv:2306.08032](#)
Zhengyan D. Shi^{*}, Shreya Vardhan^{*}, and Hong Liu
17. **Bounds on Rényi entropy growth in many-body quantum systems**
 Phys. Rev. A 109, 042404 (2024), [arxiv:2212.07444](#)
Zhengyan D. Shi
18. **Loop current fluctuations and quantum critical transport**
 SciPost Phys. 14, 113 (2023), [arxiv:2208.04328](#)
Zhengyan D. Shi, Dominic V. Else^{*}, Hart Goldman^{*}, and T. Senthil
19. **Gifts from anomalies: Exact results for Landau phase transitions in metals**
 SciPost Phys. 13, 102 (2022), [arxiv:2204.07585](#)
Zhengyan D. Shi, Hart Goldman^{*}, Dominic V. Else^{*}, and T. Senthil
20. **Many-body localization transition with correlated disorder**
 Phys. Rev. B 106, 144201 (2022), [arxiv:2204.06017](#)
Zhengyan D. Shi, Vedika Khemani, Romain Vasseur, and Sarang Gopalakrishnan
21. **Holographic flows from CFT to the Kasner universe**
 JHEP 08 (2020) 003, [arxiv:2004.01192](#)
 (α - β) Alexander Frenkel, Sean A. Hartnoll, Jorrit Kruthoff, **Zhengyan D. Shi**
22. **Topological order in matrix Ising models**
 SciPost Phys. 7 (2019) 6, 081, [arxiv:1908.07058](#)
 (α - β) Sean A. Hartnoll, Edward A. Mazenc, **Zhengyan D. Shi**

Invited Talks

- “New physics of composite fermions from symmetry reduction” Sep 2026
NUS (National University of Singapore) Condensed Matter Seminar Series ([recording](#))
- “Metallogenic quantum criticality” Sep 2026
KITP Conference: Bridging Platforms: Exploration and Control of Fractionalized States ([recording](#))
- “Charge 4e superconductivity with parafermionic vortices” Aug 2026
EPiQS Postdoctoral Symposium
- “Anyon transmutation across FCI-SC transitions” Jun 2026
Gordon Research Seminar
- “Non-Abelian topological superconductivity from melting Abelian FCIs” Mar 2026
Cornell Quantum Theory Seminar ([recording](#))
- “Non-Abelian topological superconductivity from melting Abelian FCIs” Feb 2026
Flatiron Institute Moire meeting
- “Non-Abelian topological superconductivity from melting Abelian FCIs” Feb 2026
University of Maryland Condensed Matter Theory Center Seminar
- “Anyonic Quantum Matter” Oct 2025
Geballe Laboratory for Advanced Materials: Condensed Matter Physics Seminar
- “Chiral superconductivity from non-Abelian anyon fluids” Oct 2025
Anthony J Leggett Institute for Condensed Matter Theory: Special Seminar ([recording](#))
- “Doping non-Abelian fractional quantum anomalous Hall insulators” Jun 2025
UC San Diego Condensed Matter Theory Seminar
- “Doping a fractional quantum anomalous Hall insulator” Dec 2024
Stanford Condensed Matter Theory Seminar
- “Doping a fractional quantum anomalous Hall insulator” Dec 2024
Anthony J Leggett Institute for Condensed Matter Theory: Special Seminar
- “Non-perturbative Methods for Clean Non-Fermi Liquids ” Aug 2024
HKU-UCAS Young Physicists Symposium 2024: “Non-Fermi liquids and anomalous transport in two dimensions”
- “Excitonic Quantum Criticality” May 2024
KITP Conference: Theories, Experiments and Numerics on Gapless Quantum Many-body Systems ([recording](#))
- “Excitonic Quantum Criticality” Jan 2024
Harvard Condensed Matter Theory Seminar
- “Controlled Expansion for Transport in a Class of Non-Fermi Liquids” Dec 2023
Stanford Condensed Matter Theory Seminar
- “Gifts from Anomalies: New Results on Quantum Critical Transport in Non-Fermi Liquids” Sept 2022
Harvard CMSA: Quantum Matter in Math/Physics Seminar ([recording](#))

Awards and Honors

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- Firestone Medal for Excellence in Undergraduate Research June 2020
- Awarded to top 10% of honors theses completed in that year at Stanford University.
 - David S. Levine Award June 2019
- Awarded to the member of the junior class at Stanford University who is judged to be the most outstanding physics student on the basis of academic breadth in addition to demonstrated abilities in experimental physics.

Teaching Experience

Teaching Assistant, MIT

Jan. 2021 - May. 2025

- Course: Physics 8.223 (Lagrangian Mechanics), Physics 8.333 (Statistical Mechanics I)
- Office hours, recitations, problem set grading.

Course Assistant, Stanford

Sept. 2019 - June 2020

- Courses: EE222 (Applied Quantum Mechanics I), MATH215B (Graduate Differential Topology), MATH215C (Graduate Differential Geometry)
- Graded assignments and wrote solutions to problem sets.

Volunteer Instructor, Stanford Splash

Nov. 2017

- Cotaught a course on quantum information and computation to high school students.

Services

Referee

May 2023 - Present

- Nature Communications, Physical Review Journals (PRB, PRL, PRR, PRX, PRX Quantum), Journal of High Energy Physics, SciPost

Mentor, MIT Directed Reading Program

Jan. 2022 - Jan. 2024

- Four one-month long directed reading programs with undergraduate students at MIT. Topics include non-Fermi liquid theory, foundations of quantum field theory, spin glass theory, and topological quantum field theory.

Mentor, Project Short

Jan. 2024 - Present

- Project Short is a non-profit that connects PhD students in the US with underprivileged PhD applicants who are seeking advice/feedback on their application.

References

Available upon request.